FINAL EXAM OF TOPOLOGY (FALL 2022)

1. (14 points) Let $S^1 = \{e^{i\theta} \mid 0 \le \theta < 2\pi\}$ and let $X = S^1 \times S^1$ be the torus. Fix $x_0 = (1,1) \in X$ and let $\gamma_1(t) = (e^{2\pi i t}, 1), \gamma_2(t) = (1, e^{2\pi i t}), t \in [0,1]$. Then we know that $\pi_1(X, x_0) = \mathbb{Z}\cdot < \gamma_1 > \oplus \mathbb{Z}\cdot < \gamma_2 >$. Let $H = 2\mathbb{Z}\cdot < \gamma_1 > \oplus \mathbb{Z}\cdot < \gamma_2 >$ which is a subgroup of $\pi_1(X, x_0)$. Then we know that H corresponds to a covering map $p_H : X_H \to X$. Describe X_H and p_H.

2. (12 points) Let $\Delta^2 = [v_0, v_1, v_2]$ be a 2-simplex. Let X be the identification space from Δ^2 by identifying v_0, v_1, v_2 to a single point. Compute $H_n(X)$ for any $n \ge 0$.

3. (14 points) Let $m \ge 1$ and $n \ge 2$ be two integers. Denote by X_m the space obtained from $\mathbb{R}^n$ by removing m distinct points.

(1) (8 points) Compute $H_k(X_m)$ for any $k \ge 0$.
(2) (6 points) Compute $\pi_1(X_m)$.

4. (12 points) Let $X = S^1 \times S^1$ be the torus and let $A = S^1 \times \{x_0\} \subset X$ be a circle in X. Let $B = \{(x, y, 0) \in \mathbb{R}^3 \mid x^2 + y^2 = 1\} \subset S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$ be a circle in S^2. Let Y be the identification space from $X \sqcup S^2$ by identifying the circle A with the circle B. Compute $H_n(Y)$ for any $n \ge 0$.

5. (12 points) Let S^n be the n-dimensional sphere. Let $f : S^n \to S^n$ be a continuous map such that $\deg f \ne 1$. Show that there exists $x \in S^n$ such that $f(x) = -x$.

6. (12 points) Let X be an $n-$dimensional CW complex. For $0 \le k \le n$, let $a_k(X)$ be the number of $k-$cells in X and let $\beta_k(X)$ be the $k-$th Betti number of X. Prove that there is a polynomial $Q(t)$ with integers as coefficients such that:

$$\sum_{k=0}^{n} a_k(X) t^k = \sum_{k=0}^{n} \beta_k(X) t^k + (1+t)Q(t).$$

7. (14 points) Is the following statement true or false? If true, give a proof. If false, give a counterexample and prove that what you give is indeed a counterexample. Recall that a continuous map $f : X \to Y$ is called *nullhomotopic* if it is homotopic to a constant map.

(1) (7 points) Every map from $\mathbb{R}P^2$ to $S^1 \times S^1$ is nullhomotopic.
(2) (7 points) Every map from $S^1 \times S^1$ to $\mathbb{R}P^2$ is nullhomotopic.

8. (10 points) Let $f : \mathbb{R}P^{2n-1} \to \mathbb{R}P^{2n-1}$ be a continuous map such that f has no fixed point. Compute $f_* : H_{2n-1}(\mathbb{R}P^{2n-1}) \to H_{2n-1}(\mathbb{R}P^{2n-1})$.